

Radar System using Ultrasonic sensor and Servo motor

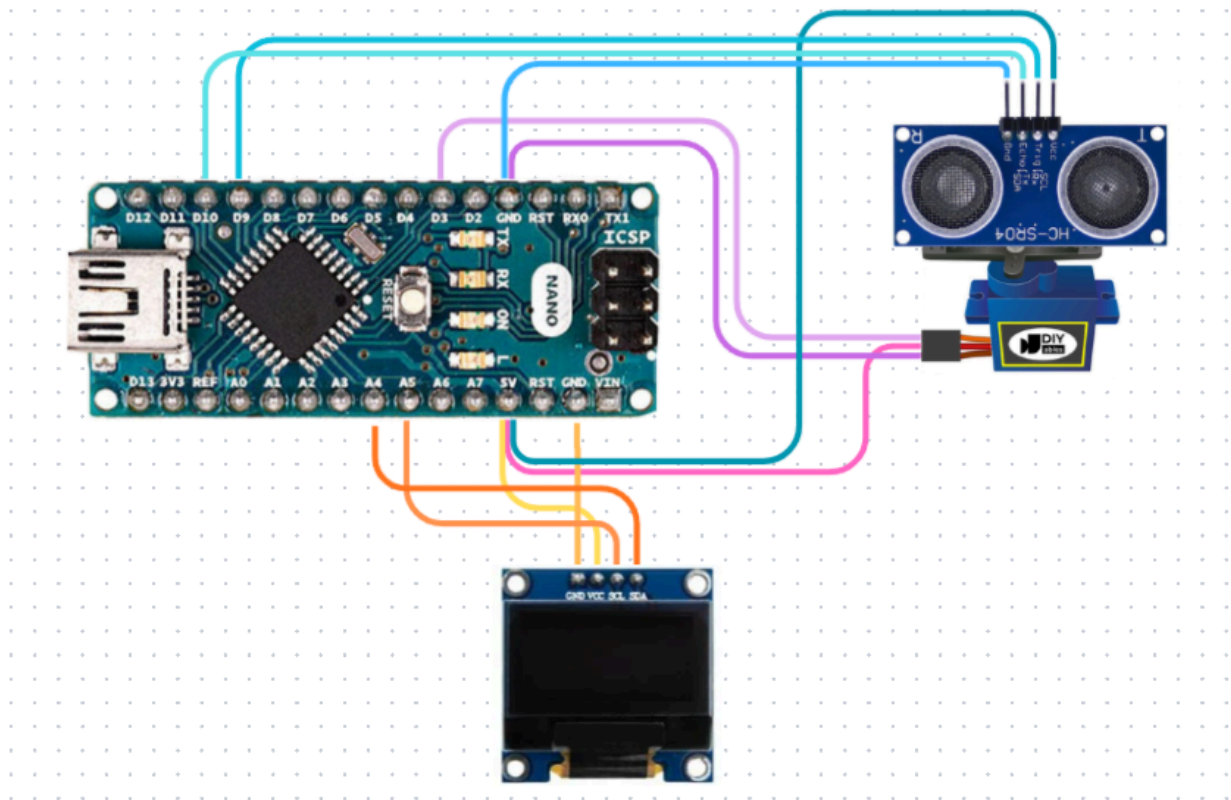
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• Components Required

Sr.No.	Component	Quantity	Description	Use
1.	Arduino Nano	1	A small microcontroller board used for programming electronic projects.	It is used to control all the components.
2.	Breadboard	1	A board with small holes used to make circuits.	It is used to make the circuit connections.
3.	0.96-inch I2C OLED Display	1	A small display used to show text and values.	It is used to show the distance detected by the sensor.
4.	Servo Motor	1	It is a motor that can move to different angles.	It is used to rotate the ultrasonic sensor.
5.	Ultrasonic Sensor	1	It is a sensor that uses sound waves to detect objects.	It is used to measure the distance of an object.

- **Circuit Diagram**



- **Program Code**

```
#include <Servo.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <math.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

Servo radarServo;

#define TRIG_PIN 9
#define ECHO_PIN 10
#define SERVO_PIN 3

#define MAX_DISTANCE 30 // Radar range (cm)
#define DETECT_DISTANCE 10 // Detection limit
```

```

// Radar center
const int cx = 64;
const int cy = 63;

float getDistance()
{
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);

    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);

    long duration = pulseIn(ECHO_PIN, HIGH, 30000);

    if(duration == 0)
        return MAX_DISTANCE + 5;

    return duration * 0.0343 / 2.0;
}

void drawRadar()
{
    display.drawCircle(cx, cy, 20, SSD1306_WHITE);
    display.drawCircle(cx, cy, 40, SSD1306_WHITE);
    display.drawCircle(cx, cy, 60, SSD1306_WHITE);

    display.drawLine(4, cy, 124, cy, SSD1306_WHITE);
}

void drawSweep(int angle)
{
    float rad = radians(angle);

    int x = cx + 60 * cos(rad);
    int y = cy - 60 * sin(rad);

    display.drawLine(cx, cy, x, y, SSD1306_WHITE);
}

void drawTarget(int angle, float distance)
{
    if(distance > DETECT_DISTANCE)

```

```

    return;

    int r = map(distance * 10, 0, MAX_DISTANCE * 10, 0, 60);

    float rad = radians(angle);

    int x = cx + r * cos(rad);
    int y = cy - r * sin(rad);

    display.fillCircle(x, y, 2, SSD1306_WHITE);
}

void drawInfo(int angle, float distance)
{
    display.fillRect(0, 0, 128, 10, SSD1306_BLACK);

    display.setTextSize(1);
    display.setTextColor(SSD1306_WHITE);

    display.setCursor(0,0);
    display.print(angle);
    display.print((char)247);

    display.setCursor(40,0);
    display.print(distance,1);
    display.print("cm");

    display.setCursor(90,0);

    if(distance <= DETECT_DISTANCE)
        display.print("OBJ");
    else
        display.print("---");
}

void scan(int startAngle, int endAngle, int step)
{
    for(int angle=startAngle;
        (step>0)? angle<=endAngle : angle>=endAngle;
        angle+=step)
    {

        radarServo.write(angle);
        delay(2);
    }
}

```

```

float distance = getDistance();

display.clearDisplay();

drawRadar();
drawSweep(angle);
drawTarget(angle, distance);
drawInfo(angle, distance);

display.display();

Serial.print(angle);
Serial.print(",");
Serial.println(distance);
}
}

void setup()
{
  Serial.begin(9600);

  radarServo.attach(SERVO_PIN);

  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);

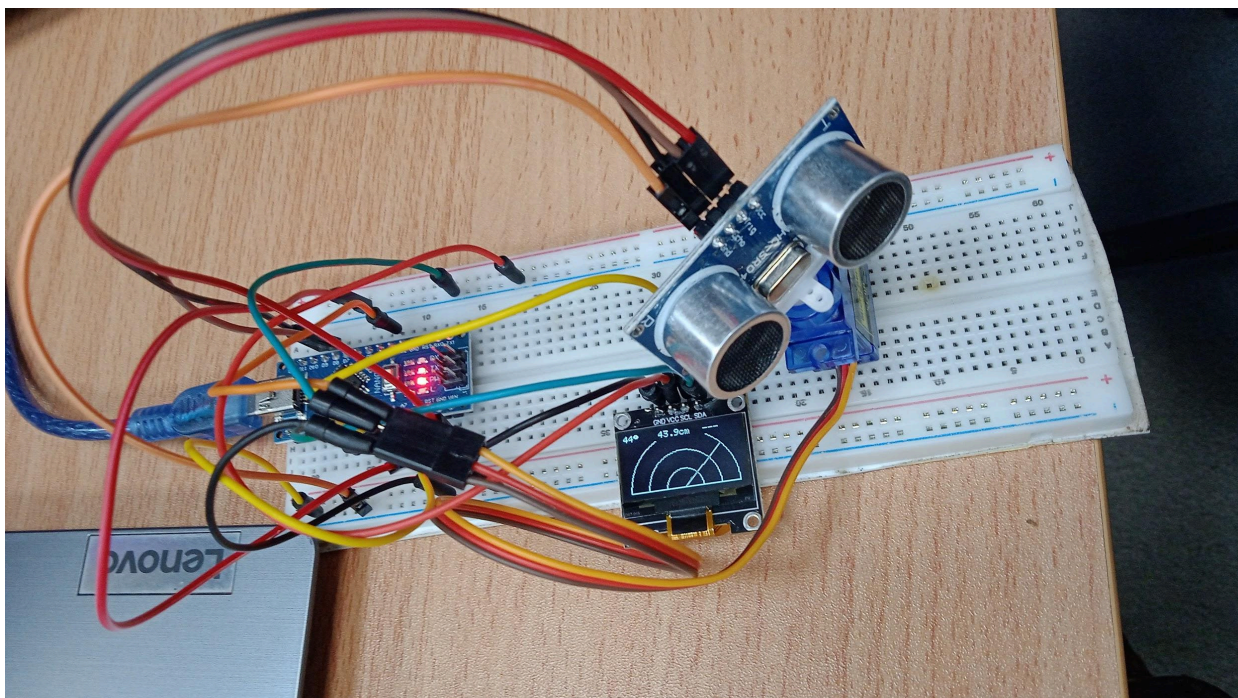
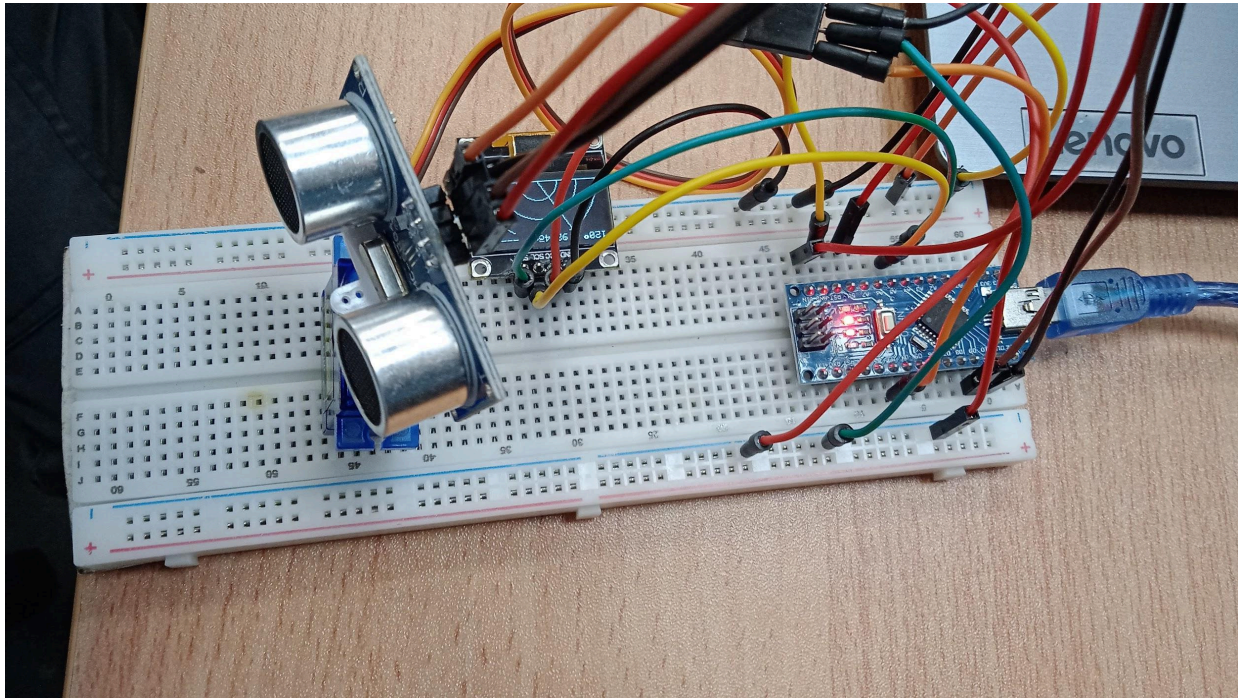
  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C))
    while(1);

  display.clearDisplay();
  display.display();
}

void loop()
{
  scan(0,180,2);
  scan(180,0,-2);
}

```


- **Photograph of the Experiment**



- **Conclusion**

In this experiment, we saw how a simple radar system works using Arduino Nano. Saw working of the ultrasonic sensor which detects objects and measures their distance and used a servo motor to rotate the sensor ,the OLED display showed the detected distance graphically.